UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

Special Right Triangles, Radians, Coordinates & Wave Projections

SERIES 1 • CARD 08 / 12 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

CARD 08 / 12 • VOLV-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

CIRCULAR COORDINATES & DYNAMIC PROJECTIONS

Trigonometry is circular motion, not triangles.

On unit circle (r = 1), coordinates are trig values:

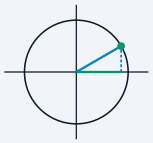
(X, Y) = (cos θ, sin θ)

Rotating disc projects angle θ onto cos/sin!

CALIBRATED 84.00 MM RADIUS DATUM

- Outer Ring: 84.00 mm ⌀ (Matches Fraction 1 Whole)
- Dual Scales: Degrees (0°–360°) and Radians (0–2π)
- Special Angles: Exact cutouts for 30°, 45°, and 60°

UNIT CIRCLE: 30° –> (cos = √3/2 ≈ 0.866, sin



ANGLE θ = 30° (π/6)

X = cos(30°) = √3/2 ≈ 0.8

Y = sin(30°) = 1/2 = 0.50

Pythagorean Identity:

cos²(θ) + sin²(θ) = 1

0.75 + 0.25 = 1.000

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC	COLOR
Unit Circle Disc	1x	84.0 mm ⌀	PLA White
Radian Overlay	1x	Radial Cursor	Clear PETG
Projection Arm	1x	Windows	Clear PC
Center Pivot	1x	Brass Bushing	Solid Brass
Stand & Lid	1x	Desk Stand	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 4 Volvelle Components + Storage Bin + Stand Lid = 6 Parts

QUADRANT SIGN CONVENTION

- Quadrant I (+, +): All functions positive (0° to 90°).
- Quadrant II (-, +): Sine positive, Cosine negative (90°–180°).
- Quadrant III (-, -): Tangent positive, Sine/Cos negative.
- Quadrant IV (+, -): Cosine positive, Sine negative.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Align transparent cursor to 0° / 360°, slip the volvelle into its protective sleeve, and lay flat into the box recess. Snap the stand lid shut for silent, rattle-free carry.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

For any real angle θ on the unit circle: cos²(θ) + sin²(θ) = 1 identically.

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LESSON GUIDE: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

Step-by-Step Hands-On Tutorial: Angle 30° -> (cos 30° = √3/2, sin 30° = 1/2)

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/unit-circle-volvelle/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS HSF.TF.A.2 (Unit Circle)
- CCSS HSF.TF.A.3 (Special Tria...)
- TEKS 111.42.c.2.P (Trig Functio...)

Grades 9–12 Trigonometry & Physics

GUIDING INQUIRY

When the cursor rotates past 90° into Quadrant II, why does the horizontal X coordinate become negative while Y stays positive?

1 STEP 1: ROTATE CURSOR TO 30° (π/6 RAD)

Turn the transparent cursor arm to the 30° mark.
Look at the inner radian scale: 30° aligns exactly with π/6 rad.
Notice the right triangle projected onto the unit circle base!

2 STEP 2: READ HORIZONTAL COSINE PROJECTION

Look down at the horizontal X-axis window:
X = cos(30°) = √3 / 2 ≈ 0.8660.
This is the horizontal base length of the 30°-60°-90° triangle.

3–4 STEPS 3 & 4: READ SINE & VERIFY ON STACKCALC

3. READ VERTICAL SINE: Look at vertical Y-axis window: Y = sin(30°) = 1/2 = 0.5000.
4. VERIFY PYTHAGOREAN IDENTITY: cos²(30°) + sin²(30°) = 0.75 + 0.25 = 1.0000!
Calculator Verification: Press Unit Circle Radian Icon ([COS], [SIN] & [⁻¹][•y,x]) and run 45 [COS] [⁻¹] [x²] 45 [SIN] [⁻¹] [x²] [+] → 1.

5 CORE PRINCIPLE: SINE AND COSINE ARE CIRCLE COORDINATES

Textbook right triangles only work for angles < 90°. The volvelle proves that sine and cosine
are the literal (X, Y) coordinates of a point moving around a circle! In Quadrant II, cosine naturally becomes negative because X is negative, while sine remains positive!

★ EXTENSION INVESTIGATION: SPECIAL ANGLE: 60° (π/3 RAD)

Rotate cursor to 60°. Notice X and Y coordinates swap: X = cos(60°) = 1/2 = 0.500, Y = sin(60°) = √3/2 ≈ 0.866! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

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